# British Auction RFQ System Assignment

**GitHub Repo Link:** **https://github.com/venkat-vamsi/British_Auction_RFQ**
(contains both frontend and backend code)

## High Level Design

## 1.Summary

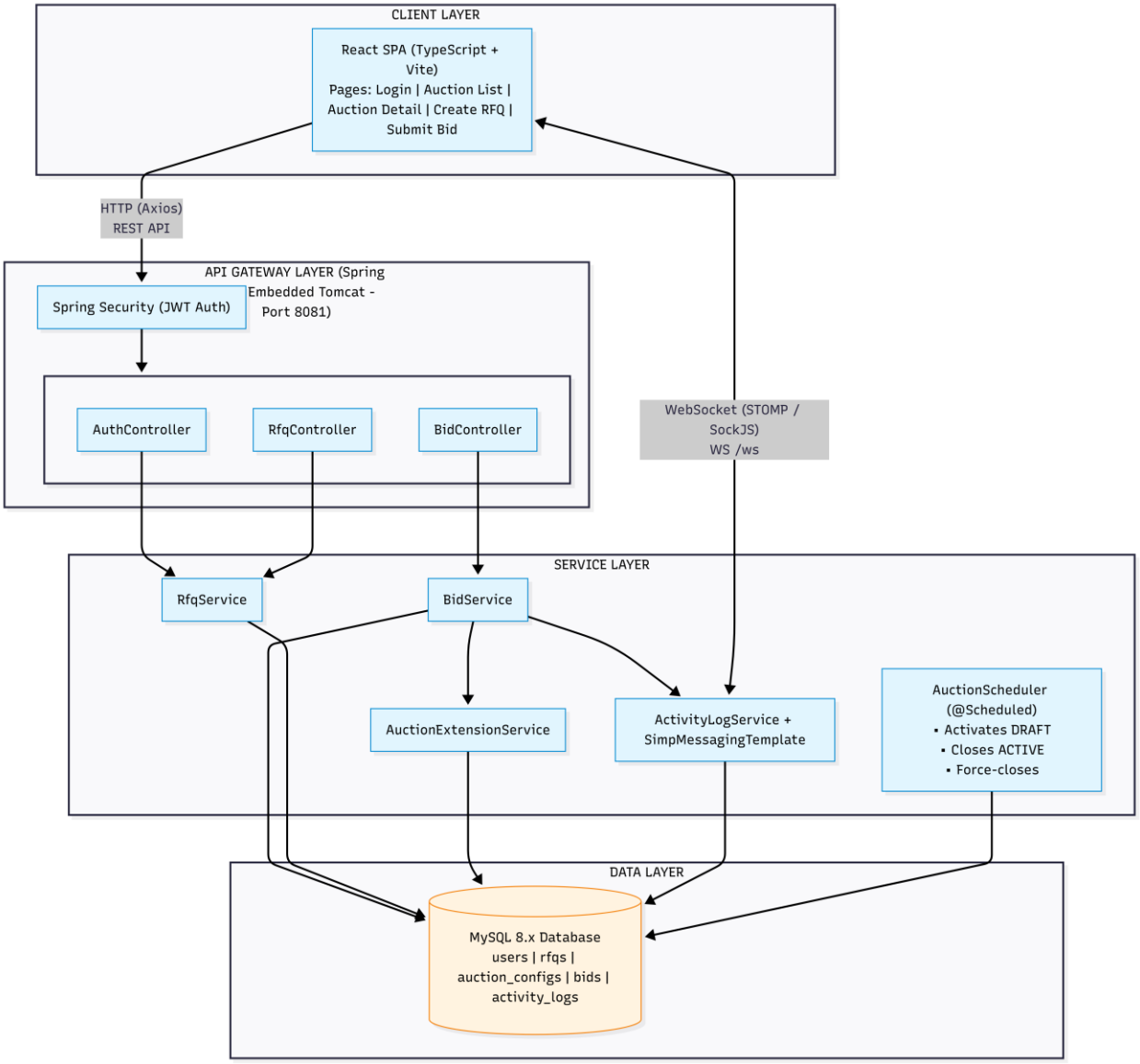
The proposed system is a Request for Quotation (RFQ) platform integrated with a British Auction-style bidding mechanism. Buyers publish logistics/freight RFQs, and suppliers compete by submitting openly visible, progressively lower bids. To prevent last-minute bidding manipulation and encourage fair competition, the system features automatic bid-time extensions triggered by late activity, capped by a mandatory forced close time.

## 3. Technology Stack

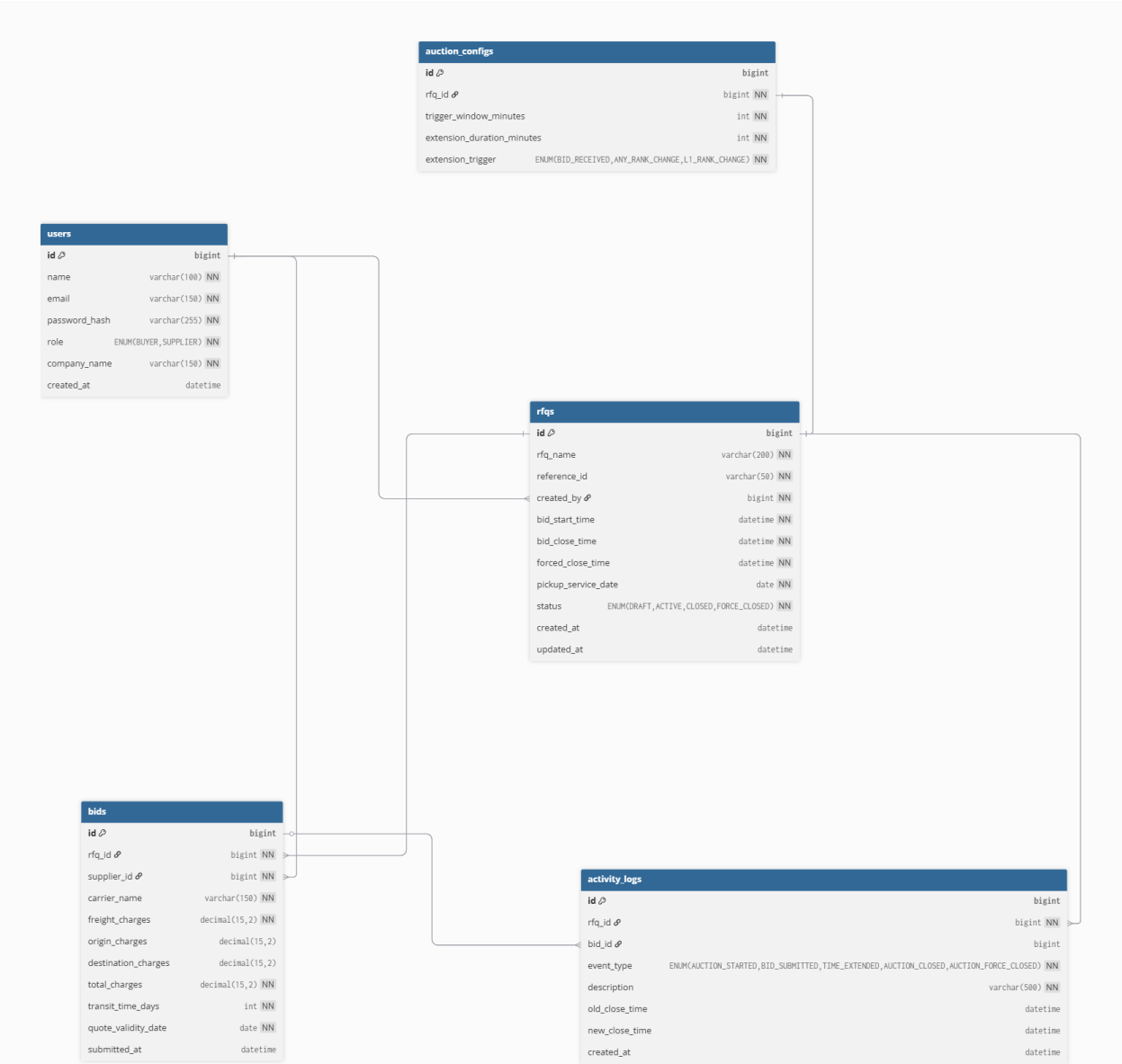
| Layer | Technology | Justification |
|---|---|---|
| **Frontend** | React 18, TypeScript, Tailwind CSS | Fast builds, type-safety, and rapid responsive styling. |
| **Real-time** | STOMP over SockJS | Reliable WebSocket communication with fallback options. |
| **Backend** | Spring Boot 3.2 (Java 17) | Production-grade REST framework with built-in scheduling (@Scheduled). |
| **Security** | Spring Security + JWT | Stateless authentication suitable for scalable APIs. |
| **Database** | MySQL 8.x, Spring Data JPA | ACID compliance and reduced boilerplate for relational data mapping. |

## 2. System Architecture

The system utilizes a modern, decoupled client-server architecture, leveraging WebSockets for the real-time updates critical to a live auction environment.



# 4.SCHEME DESIGN



# 5. Core Business Logic & Auction Configurations

The ‘AuctionExtensionService' manages the core rules of the British Auction. These configurations dictate when and how the auction time is extended.

- **Trigger Window (X Minutes):** Defines the monitoring window just before the auction ends.
- **Extension Duration (Y Minutes):** The amount of extra time added when a trigger occurs.
- **Extension Triggers:** The system supports three specific triggers:
    - **Bid Received:** Any new bid during the trigger window extends the auction.
    - **Any Rank Change:** Any change in the overall supplier ranking extends the auction.
    - **L1 Rank Change:** The auction only extends if the lowest-priced supplier (L1) changes.
- **Strict Validations:** The Forced Bid Close Time must always be greater than the standard Bid Close Time. Any time extensions applied by the ‘AuctionExtensionService’ must never exceed the Forced Bid Close Time.

# 6. Component Breakdown

### 6.1 Frontend Interface

- **Auction Listing Page:** Displays all active and closed British Auctions, summarizing the RFQ ID, Current Lowest Bid, Close Times, and overall Status.
- **Auction Details Page:** Displays a live feed of all supplier bids sorted by price, supplier rankings (L1, L2, etc.), detailed quote data, the specific X/Y configurations, and a real-time activity log detailing bid submissions and the reasons for any time extensions.

### 6.2 Backend Services

- **BidService:** Validates that the auction is active, ensures new bids are lower than the supplier's previous best, computes pre- and post-bid rankings, persists data, and triggers the extension check.
- **AuctionExtensionService:** Evaluates the current time against the Trigger Window (X). If a valid trigger condition is met, it calculates the new close time (Current + Y), ensuring it does not bypass the forced close time, and broadcasts updates via WebSocket.
- **AuctionScheduler:** A background process running at intervals to automatically transition RFQ states (e.g., Draft to Active, Active to Closed, or triggering the Forced Close).

## 7. Ranking Algorithm

Rankings are computed dynamically (not stored). For each RFQ:

1. Group all bids by `supplier_id`
2. For each supplier, select the bid with the lowest `total_charges`
3. Sort suppliers by `total_charges` ASC
4. Assign rank: L1 (lowest), L2, L3, ...

## 8. API Summary

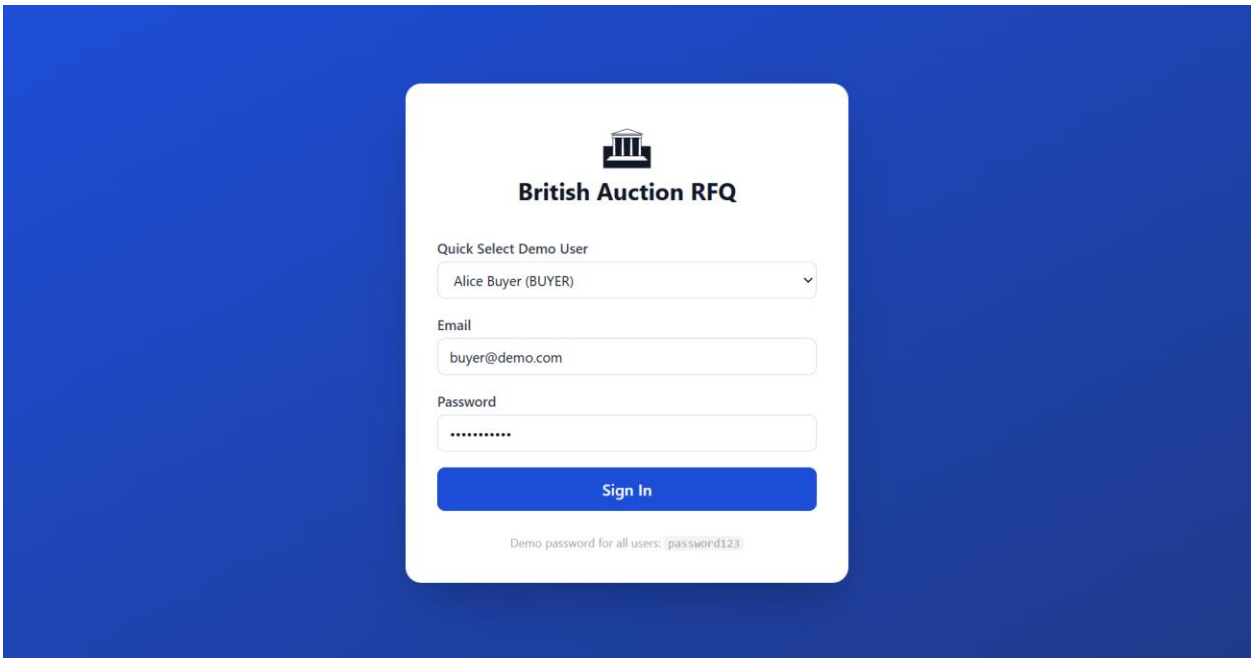
- POST /api/auth/login → Login, get JWT token
- GET /api/rfq → List all RFQs (summary)
- POST /api/rfq → Create RFQ (BUYER only)
- GET /api/rfq/{id} → Get RFQ full detail
- POST /api/rfq/{id}/bids → Submit bid (SUPPLIER only)
- GET /api/rfq/{id}/bids → Get all bids with rankings
- GET /api/rfq/{id}/activity-log → Get activity log
- WS /ws → WebSocket endpoint
- TOPIC /topic/auction/{id} → Live auction updates

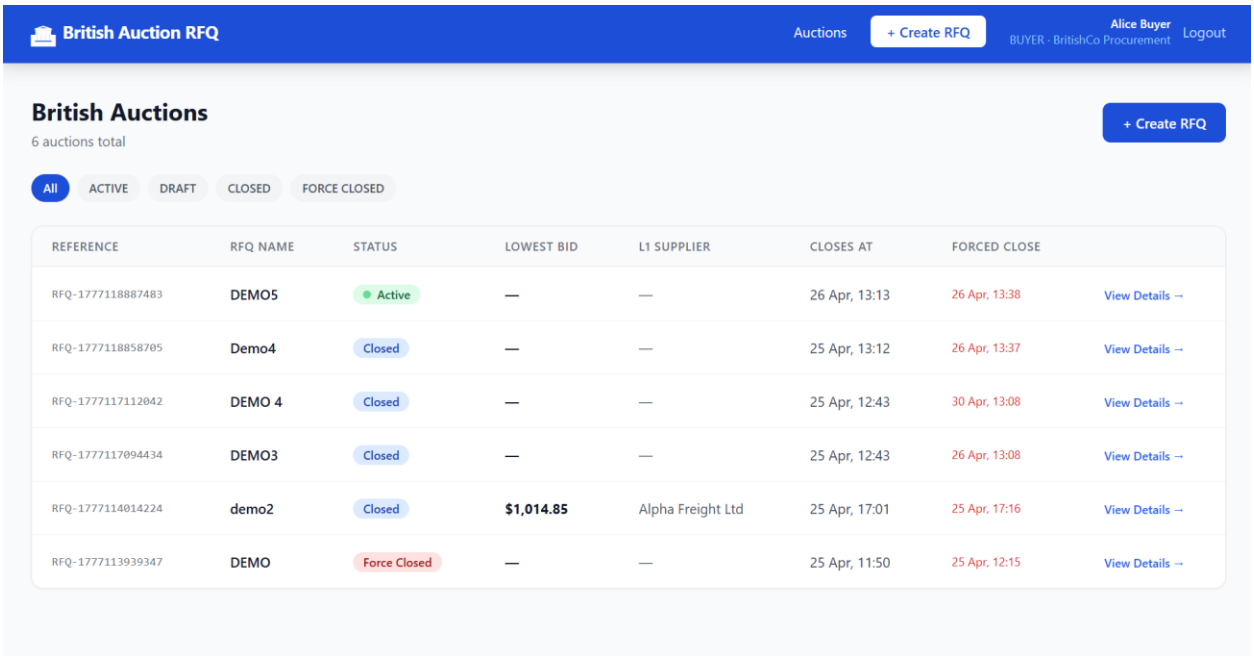
## 9. Real-Time Data Flow

To ensure suppliers have immediate visibility of being outbid, the system utilizes a unidirectional real-time broadcast:

1. Supplier submits a bid via POST /api/rfq/{id}/bids.
2. BidService validates and saves the bid to the MySQL database.
3. AuctionExtensionService evaluates if an extension is required based on the configured triggers.
4. ActivityLogService records the bid and any resulting time extensions.
5. SimpMessagingTemplate broadcasts the updated state (new bid, new rankings, extended time) to /topic/auction/{id}.
6. Subscribed React clients instantly re-render the auction table, rankings, and countdown timer.

# 10. Website images:



← All Auctions

# DEMO5

Ref: RFQ-1777118887483 · Posted by BritishCo Procurement

Active | Submit Bid

BID CLOSES IN

**19:31:04**

Forced close: 26/4/2026, 1:38:00 pm

**Auction Details**

| | |
|---|---|
| Bid Start | 25 Apr 2026, 12:13 |
| Bid Close | 26 Apr 2026, 13:13 |
| Forced Close | 26 Apr 2026, 13:38 |
| Service Date | 02 May 2026 |

**British Auction Config**

| | |
|---|---|
| Trigger Window | 10 min before close |
| Extension Duration | +5 min |
| Extension Trigger | Any rank change |

Ranked Bids (3) | Activity Log (3)

| RANK | SUPPLIER | CARRIER | FREIGHT | ORIGIN | DESTINATION | TOTAL | TRANSIT | VALID |
|---|---|---|---|---|---|---|---|---|
| L1 | Beta Logistics | ABX | $700.00 | $18.00 | $50.00 | **$768.00** | 15d | 01 Ja 2028 |
| L2 | Alpha Freight Ltd | XYZ | $1,000.00 | $18.00 | $120.00 | **$1,138.00** | 12d | 01 Ja 2027 |
| L3 | Gamma Cargo Co | PQR | $900.00 | $100.00 | $173.89 | **$1,173.89** | 12d | 01 Ja 2029 |

← Back to Auction

# Submit Bid

DEMO5 Active

Closes At: 26 Apr, 13:13:00

Forced Close: 26 Apr, 13:38:00

**Quote Details**

Carrier Name *

e.g. DHL Express

Freight Charges (USD) *: 0.00

Origin Charges (USD): 0

Destination Charges (USD): 0

Total Quote: **$0.00**

Transit Time (days) *: e.g. 5

Quote Validity Date *: dd-mm-yyyy

Cancel | Submit Bid

BID CLOSES IN

**19:21:12**

Forced close: 26/4/2026, 1:38:00 pm

### Auction Details

| | |
|---|---|
| Bid Start | 25 Apr 2026, 12:13 |
| Bid Close | 26 Apr 2026, 13:13 |
| Forced Close | 26 Apr 2026, 13:38 |
| Service Date | 02 May 2026 |

### British Auction Config

| | |
|---|---|
| Trigger Window | 10 min before close |
| Extension Duration | +5 min |
| Extension Trigger | Any rank change |

### Summary

| Total Bids | Lowest Bid |
|---|---|
| 3 | $768 |

Ranked Bids (3) | Activity Log (3)

- Bid submitted by Gamma Cargo Co — Total: 1173.89 — 17:41:28
- Bid submitted by Beta Logistics — Total: 768.00 — 17:40:36
- Bid submitted by Alpha Freight Ltd — Total: 1138.00 — 17:39:55

British Auction RFQ

Auctions | + Create RFQ | Alice Buyer BUYER · BritishCo Procurement | Logout

← Back

# Create RFQ

Configure a British Auction for your freight request

## RFQ Information

RFQ Name *

e.g. London to Manchester Freight Q2

Reference ID *

RFQ-1777119369075

Pickup / Service Date *

02-05-2026

## Auction Timing

Bid Start Date & Time *

25-04-2026 12:21

Bid Close Date & Time *

Auction will try to close at this time (may extend)

25-04-2026 13:21

Forced Close Date & Time *

Hard deadline — auction NEVER extends beyond this

25-04-2026 13:46

## British Auction Configuration

Trigger Window (X minutes)

Monitor bidding activity this many minutes before close

10

Extension Duration (Y minutes)

Add this many minutes when trigger condition fires

5

Extension Trigger

- (●) **Any Bid Received**
  Extend whenever any supplier submits a bid in the trigger window
- ( ) **Any Rank Change**
  Extend when any supplier's ranking position changes
- ( ) **L1 Rank Change**
  Extend only when the lowest bidder (L1) changes

⚙ Auction Logic Preview

The system will monitor bidding in the last **10 minutes** before close. If triggered (any bid received), the auction extends by **5 minutes**, but never beyond the forced close time.

Cancel | Create RFQ